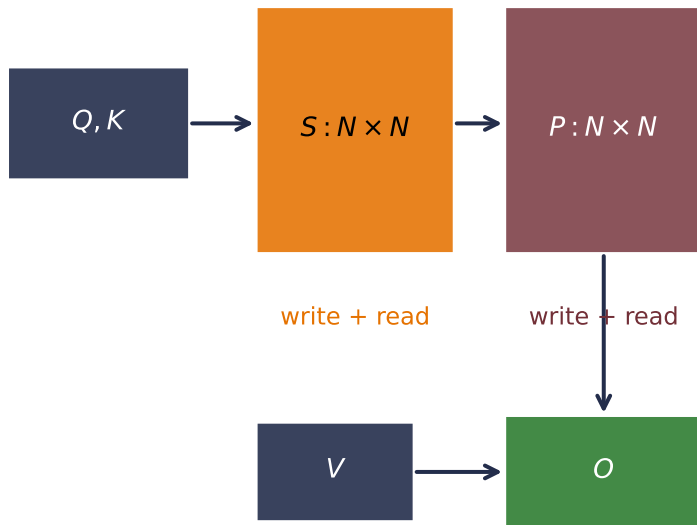
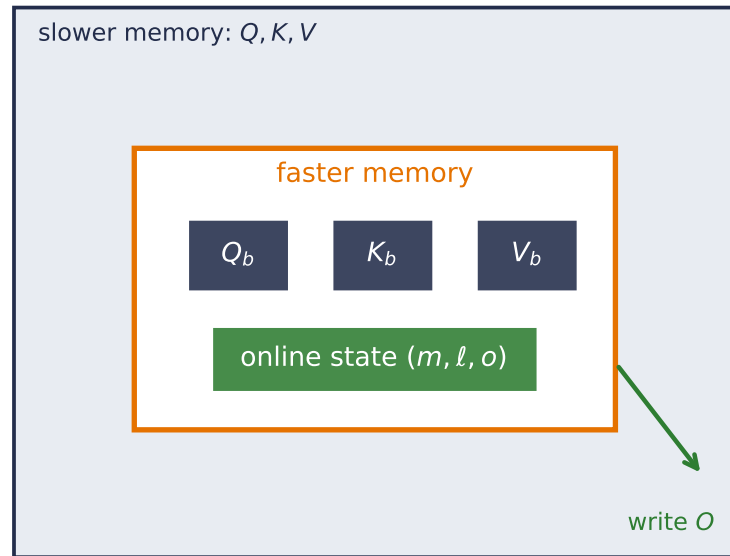


materialize quadratic intermediates



slow-memory traffic includes full grids

tile and normalize online



no full score or probability grid written